SSHVisual: Visual Tool to Analyze SSH Data

Shreeman Gautam
Kahlert School of Computing
University of Utah
Salt Lake City, UT, USA
email address or ORCID

Robert Ricci
Kahlert School of Computing
University of Utah
Salt Lake City, UT, USA
email address or ORCID

***Abstract*—Network operators use visualizations to detect attacks on their systems, interacting with data to uncover patterns that might stay hidden. Operators use these findings to block attackers and to expand the knowledge base of known threats. Detecting anomalies and identifying relationships iteratively in data is, at best, tedious and, at worst, infeasible without data visualizations. Prior work in network visualization analyzed the IP address and ASN dimensions of SSH logins with little focus on the username aspect of the attackers' attempts and detected anomalous network events when baseline events were available. As a result, little attention has been paid to whether attackers that differ in timing, source address, and username belong to the same coordinated campaign. We present SSHVisual, an iterative visualization system evaluated on login attempts collected from our production SSH systems. SSHVisual analyzes the IP, ASN, and username dimensions of SSH attacks, to support campaign detection. It complements automated detection by allowing operators to find anomalous events without a normal baseline. It uses a similarity search across dimensions that groups attackers by behavior and explains each match. It supports iterative and collaborative sharing of findings between operators. We evaluated SSHVisual on 206.6 million attacker login attempts collected over 69 days. We found that it helps operators surface hidden patterns and find attackers to block. In one case study, some IPs tried a similar set of usernames, including uncommon ones, which pointed to a single obscure campaign that volume charts could present as unrelated bursts. In another, similarity search surfaced low-volume IPs working through various username lists, none of which ranked in the top ten by volume. Without similarity search, these IPs would likely have been hard to find and block. These results suggest that username-centric, operator-driven exploration can be valuable because it can reveal covertness and coordination that previous analyses can miss.**

***Index Terms*—SSH, brute-force attacks, attack campaigns, network security visualization, visual analytics, similarity search**

I. Introduction

Computer systems employ SSH [1] to allow users to securely connect to them in order to execute commands and do remote administration. Many users do a username, password combination (password based authentication) in order to gain entry to the SSH system. Attackers, who are adversarial, can use brute force attack mechanisms [2] to get into SSH systems. Sometimes, they guess the username, password combinations and other times, they take advantage of poorly secured systems to build botnets [2] to carry out further attacks [3], [4] or steal resources from the systems, like cryptomining [5], [6].

Given the prevalence of SSH attacks in the IPv4 space [2], we designed SSHVisual and evaluated it on login attempts from our production SSH systems. SSHVisual serves two kinds of network operators: Network administrators, who handle daily SSH traffic, need to identify attackers quickly and block them and Security analysts, who study traffic over longer periods, look for patterns that are not apparent and share what they find. Consider a single period: an administrator can isolate the highest-volume attacker and block it, while an analyst examining the same window can iteratively surface a set of low-frequency attackers that attempt an identical sequence of usernames, and group them as a coordinated campaign.

Surveys of network visualization [7]–[11] establish the maturity of the field and the gaps we target. Our two contributions follow from treating the username, alongside the source IP and ASN, as part of an attacker's identity, and each one serves one of the two operators described above. First, for security analysts, we show that username-based similarity identifies coordinated campaigns that volume-based views split apart. IPs that try the same list of usernames can be recognized as a single campaign, even when they come from different countries and ASNs and are never active on the same day. Second, for network administrators, we show that behavioral similarity search finds attackers that would otherwise stay hidden. Starting from a single IP the administrator already intends to block, it surfaces low-volume IPs that act alike but never reach the top-ten volume rankings, and it explains each match, so the administrator knows which other IPs to block and why. Since network traffic may not offer a stable, attack-free baseline of normal behaviour [12], detectors that rely on such a baseline may miss activity that doesn't deviate from it in the expected way. Operators, by contrast, may recognize coordination and similarity across sessions visually, without specifying normality in advance. The above contributions build on this ability and are meant to complement baseline-driven detection. As operators explore, they can label and share findings so colleagues do not investigate the same patterns twice, and the findings can in turn inform automated detection.

The rest of the paper is as follows: Section II is Related Work, Section III is Tasks and Requirements, Section IV is Design, Section V is Evaluation and Section VI is Conclusion.

II. Related Work

A long line of visualization tools [13]–[22] renders network traffic and security events navigable for operators, the majority being network-flow and traffic oriented [13], [14], [16],

[18], [19], [21], [22], with some also integrating intrusion-detection/prevention and host logs [15], [17] or multidimensional cyber-defense data [20]. Additionally, [23] visualizes IDS alerts and [24], [25] show heterogeneous security logs. [26] extract multi-resolution discrete-wavelet features from traffic, project them with principal component analysis, and apply hierarchical clustering across several intrusion-detection benchmarks, separating brute-force attacks into distinct clusters. Related work on coordinated multiple views [27]–[29] and insight generation [30] informs our design.

A. Username Campaigns with Time

To our knowledge, honeypot literature has rarely treated the username identity as an actor for campaign detection purposes. Typically, it is a dimension to count rather than to track, whereas in our analysis it serves, alongside the IP and ASN, as an anchoring dimension. Several papers [31]–[37] collect or chart honeypot traffic, with several of them capturing SSH credentials (usernames and passwords). Where credentials are charted, like the SCADA SSH honeypot [34], kippo-graph [35], and the container deployment [36], usernames and passwords appear as static top-N tallies (kippo-graph's top-10 username and password bars being an example), and not placed on a time axis. The WAN honeypot framework [38] is the closest prior system : it computes an IP-to-IP dictionary-overlap (containment) score to probe whether six same-/24 source IPs share a wordlist, which is a manual, one-off prototype of credential-based coordination analysis, performed on the IP and without a temporal dimension.

B. Username-Based Similarity

Computed similarity between entities is well established in literature [39]–[42]. We do not claim similarity computation itself as novel; the contribution lies in the entity it operates on. Prior systems cluster internal hosts [39], behavioral profiles [41], IP/port members [42], or internal enterprise users grouped by shared resources [40]. None treat the username as an attacker entity. We use the username as a signature of an IP, grouping IPs by the credentials they try, with uncommon usernames weighted as stronger evidence than ones nearly every attacker tries. Honeypot/honeynet analyses have related attack activity through time by grouping attacks whose time series rise and fall together [43], [44]. Attackers that share a credential list, however, need not be active together, so we compute similarity independently of when each attacker is active and let the operator see their time series in a linechart.

C. Limits of Automated Detection

Automated anomaly detection assumes a normal baseline, and SSH traffic generally does not have one. Even where a baseline exists, such detection is fragile. [45] find that machine-learning anomaly detection is rarely deployed, because normal traffic varies widely, errors are costly, and sound evaluation is hard. [46] show that when intrusions are rare, the false-alarm rate limits any intrusion detector. Our data is the opposite case, since nearly every login attempt is an attack. [12] question whether attacks are rare and whether attack-free training data exists at all, and note that malicious traffic can look normal. [39] and [47] learn a model of each actor's expected behavior with [47] assuming that an actor's activity repeats weekly. However, constantly churning, near-uniformly hostile traffic like we see cannot offer a per-actor normal. Situ also concedes that operators "do not want another automated tool with algorithms they do not trust" and keeps a human in the loop. Lacking a per-actor baseline, we make campaign identification operator-driven from the start.

D. Username as Actor Identity

We analyze attackers at the level of a campaign, meaning a group of attackers acting together, rather than one attacker at a time. That raises a practical problem: a campaign rarely shows up as a single source IP. IPs may not be tied to one machine. One IP can be assigned to different hosts [48], hosts can cycle through new IPs [49], and counting by address can overstate the number of real machines sometimes by an order of a magnitude [50]. A view organized by address is therefore likely to scatter one campaign across dozens of entries that can look unrelated. The username can pull those pieces back together. Attackers in the same campaign likely draw from a shared credential list, so the same login names may recur across changing IPs. Grouping by username can reassemble the campaign that an address-by-address view could split apart.

E. Operator Labeling and Collaboration

The findings an operator produces can be captured and shared. In two works [19], [51], interactive labeling was used to fold the operator's judgments back into the system. Collaborative analysis was supported through shared artifacts with [17] and [20] being examples. These systems establish that labeling and collaboration are valuable, which is why our labeling-and-sharing mechanism is the step that turns an identified set of usernames into a durable, shared campaign.

III. Tasks and Requirements

The design of SSHVisual draws on the authors' experience as network operators and on gaps we identified in prior work on SSH attack analysis and security visualization. From these we derive three tasks and five requirements.

A. Tasks

T1: Identify SSH campaigns. Security analysts need to find groups of attackers that behave alike across usernames, IP addresses, autonomous systems (ASes), and countries, and to record those groups so they can be referenced later.

T2: Detect anomalies that warrant action. Network administrators need to notice unusual activity, such as a sudden spike or a single source dominating traffic, quickly enough to decide whether to block it.

T3: Share findings across operators. Analysts and administrators need to see what their colleagues have already investigated, so that the same patterns are not identified twice.

B. Requirements

R1: Attacker similarity profiling. The system should be able to cluster similar attackers based primarily along the username dimension, secondarily along IP and AS information and tertiarily along other dimensions. R1 mostly serves T1 as it allows analysts to build a repository of similar attackers.

R2: Sorting and ranking. SSH data are heavy-tailed: a few usernames [52] and IP addresses [53] account for most attempts, and the remaining entities are buried beneath them. The system should therefore list every entity in each dimension, not only the most active ones, and let the operator sort and rank entities by any combination of attributes. R2 supports T1 and T2: administrators can surface anomalous entities that never reach the top of a volume ranking, and analysts can find entities with similar attribute values.

R3: Shared tagging. Operators should be able to tag attackers, record both what a tag means and why it was applied to a particular attacker, and have those tags persist and remain visible to every operator using the system. Each tag application should record who applied it and when, so that later operators can judge whether a finding is still current. R3 primarily supports T3, and it supports T1 because tags accumulate into a repository of related attackers.

R4: Complementary tabular and temporal views. No single visualization serves every need. A table supports precise lookup and ranking but discards temporal structure; a linechart preserves temporal structure, showing spikes and campaigns at a glance, but conveys values less precisely. The system should provide both views and let operators move between them. An administrator may find and act on an anomaly using the table alone, whereas an analyst may use the table to assemble a set of related attackers and then examine their behavior over time in the linecharts. R4 supports T1 and T2.

R5: Interactive exploration of the linecharts. The linecharts should follow Shneiderman's Visual Information-Seeking Mantra: overview first, zoom and filter, then details-on-demand [54]. They should open with an overview of the full time range, let the operator zoom into a time window and filter which entities are displayed, and reveal details about an individual entity on request. R5 supports T1 and T2: zooming isolates the time window of a suspected campaign, and filtering removes dominant entities that would otherwise mask smaller ones. Additionally, line charts become harder to read accurately as more series are added [55]. Each linechart should therefore show at most ten entities.

IV. Design

SSHVisual combines two visual components, a discovery table and a set of linecharts, with two non-visual components, similarity search and shared tagging. We describe the visual design in Section IV-A and the non-visual design in Section IV-B. All figures show SSH login attempts collected between 1 November 2022 and 8 January 2023, a period of 69 days.

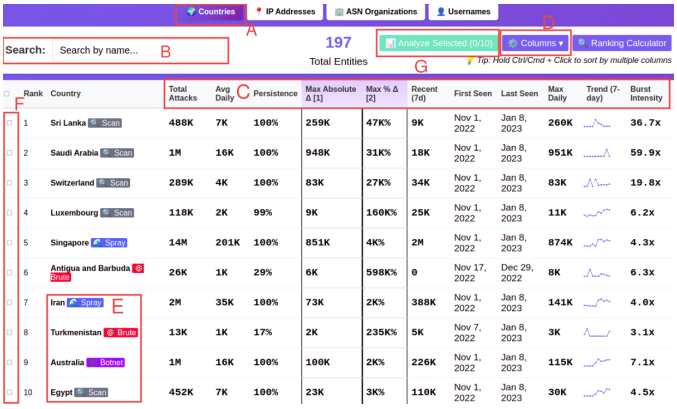


Fig. 1. Discovery table, showing the countries tab (A). Operator can filter any country by name (B). Each row shows a country and operators can sort by any mix of columns (C), with additional columns present in (D). Based on attack patterns, a tag is assigned (E). Operator can select up to ten rows (F and G) to plot in linechart.

A. Visual Design

The interface provides two views of the data. The discovery table (Fig. 1) is time independent: each row summarizes one entity over time. The linecharts (Fig. 2) can be time-dependent. Both views cover the full data period by default.

1) Discovery Table: The discovery table has four tabs, one per dimension: Countries, IP addresses, ASN organizations, and Usernames. In Fig. 1 the Countries tab is active (label A) and the other tabs are inactive. A search box filters rows by name (label B), and the counter above the table reports the number of entities in the current tab (197 countries). Some columns (label C) are shared across all tabs: total attacks, average daily attacks, persistence (the share of days on which the entity was active), the largest absolute and percentage day-over-day change, attacks in the most recent seven days, first- and last-seen dates, maximum daily count, trend (mini line chart showing attacks at 7-day intervals) and burst intensity (ratio of max daily attacks to average daily attacks where a higher ratio likely indicates an entity that attacks sporadically but with higher frequencies).

Exclusive to the countries tab, each country entity also carries a badge summarizing its dominant behavior, namely Scan, Spray, Botnet or Brute (label E). Brute marks entities whose traffic comes from few IP addresses making many attempts on few usernames. Spray marks traffic from many IP addresses trying many usernames, and Botnet marks traffic spread across many IP addresses and many ASNs. Scan marks a general attack pattern that fits none of the other three.

Unlike the linecharts, which show only the ten most active entities per dimension, the table lists every entity. Operators can select up to ten rows with the checkboxes (label F) and use Analyze Selected (label G) to plot those entities in the linecharts, and they can return to the linecharts at any time. This link lets operators move from a precise, ranked lookup to the temporal behavior of the same entities, fulfilling R4.

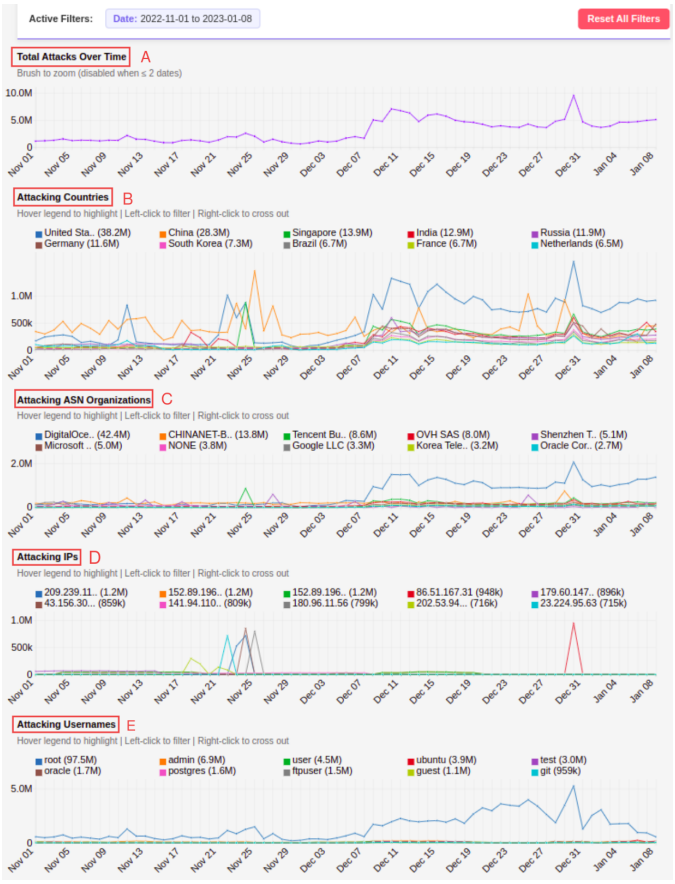


Fig. 2. The aggregate chart (A) at the top shows total attacks over time; the four charts below show the ten most active Countries (B), ASNs (C), IPs (D), and Usernames (E).

2) Linecharts: The linechart view (Fig. 2) stacks five charts on a shared daily time axis. The first, Total Attacks Over Time, shows the aggregate volume of attempts; the remaining four show the ten most active countries, ASNs, IP addresses, and usernames. Each legend entry lists the entity's total for the selected period. Interaction with the linecharts follows Shneiderman's mantra [54] (R5), and the charts are coordinated so that an action in any one of them updates all of them.

Overview first. By default, the charts cover all sixty nine days of data, with each per-entity chart showing the ten most active entities in its dimension.

Zoom. Every chart supports brushing. When the operator brushes a time range on any chart, the range is applied to all five charts and recorded in the Active Filters bar, so the temporal context stays the same whichever dimension the operator is examining. Brushing is disabled once the range is two days or fewer, since a line needs more points to show a trend.

Filter. Right-clicking a legend entry crosses the entity out, removing it from its chart and rescaling the y-axis so that smaller series are no longer flattened against the axis by a dominant one. Left-clicking a legend entry selects that entity and filters every chart to the attacks associated with it. Because each attack attempt links a username, an IP, an AS, and a country, the selection propagates along these relationships:

- Selecting a country restricts the AS, IP, and username charts to the entities observed in attacks from that country
- Selecting an AS shows the country it originates from, along with its IP addresses and the usernames they try
- Selecting an IP address shows its country and AS, along with the usernames it tries
- Selecting a username shows the countries, ASes, and IP addresses from which it was tried

Details-on-demand. Hovering over a data point shows a tooltip with that entity's attack count for the day, hovering over a legend entry highlights the entity's line, and the legend reports each entity's total for the selected period.

Linked filtering turns the linecharts from four separate rankings into a means of tracing an entity across dimensions. An operator investigating the spike on 30 December, for example, can select `root` in the username chart; the IP, AS, and country charts then show which addresses, networks, and countries tried `root`, and brushing around 30 December narrows the view to the spike itself. Conversely, selecting an AS such as DigitalOcean reveals which usernames its hosts try, indicating whether traffic from a hosting provider reflects one campaign or many. Brushing on every chart, linked filtering, and on-demand details together fulfill R5.

The linecharts answer questions that the discovery table cannot, such as "Do the countries in Fig. 1 share the same peaks and valleys?" or "Does one country begin attacking earlier than another?" In Fig. 2, the aggregate chart shows a sharp spike on 30 December, and most of the top ten countries peak on the same day, indicating a coordinated, multi-country event rather than activity from a single source. The charts also reveal differences in timing. China's largest peak falls in late November, weeks before the US' activity rises, and the IP chart shows individual addresses that each spike for a day or two and then fall silent. Neither pattern can be recovered from the table's per-entity totals. Together, Figs. 1 and 2 show how the two views complement each other, fulfilling R4.

3) Sorting and Ranking: Fig. 3 shows the usernames tab sorted on two columns (labels A and B). Operators add sort keys by holding Ctrl/Cmd while clicking column headers, and the bracketed indices in the headers show when the column was clicked.

Sorting matters because the data are heavy-tailed. The username linechart in Fig. 2 is dominated by `root`, which accounts for tens of millions of attempts and flattens every other line. Ranking by day-over-day change instead surfaces usernames (label C in Fig. 3), which are not seen in Fig. 2, several of which are default accounts for specific software or hardware products (`karaf`, for example, is the default account of Apache Karaf). Their totals are far below the top ten, but their activity is concentrated: `karaf` received 77.6K attempts in total, 44.8K of them on a single day, and was active on only 15 of the 69 days. Bursts of product-specific credentials like these suggest campaigns targeting particular devices, yet

they are invisible in a volume-ranked chart. Fig. 3 thus shows how sorting and ranking over the full entity list fulfills R2.

B. Grouping and Tagging

The visual components help operators find individual attackers, while the grouping and tagging components help them connect those attackers to others and record what they find. Similarity search groups attackers by behavior, in the username dimension (Section IV-B1) and the IP dimension (Section IV-B2), and shared tagging (Section IV-B3) preserves the resulting groups so that other operators can build on them.

1) Username Similarity: Similarity search lets operators start from one attacker and retrieve the attackers that behave most like it. For usernames, the unit of analysis is the population of sources that tries a given username. For each username we compute the breadth of that population (the number of distinct IP addresses, ASes, and countries that try it), its volume, its persistence, and how concentrated its traffic is on its single most active source. We do not reuse the seventeen-dimensional profile we apply to IP addresses (Section IV-B2), because a diagnostic analysis showed several of those features to be uninformative for usernames. Day-to-day stability is the clearest example: most IP addresses are active on many days, so their stability is well defined, but about half of the usernames appear on only a single day, where stability is meaningless. We cluster the username profiles with k-means, selecting $k = 9$. Each cluster is labeled automatically from its centroid. To answer a query, the system retrieves the query's cluster and ranks the other members by Euclidean distance to the query, reported as a similarity score.

In Fig. 4, an operator searches for the username `james` (label A). The system places it in Cluster 3, labeled High-Volume Persistent Botnet-Scale, which contains 2,624 usernames; the cluster summary reports that a member receives about 31.8K attempts from roughly 4K IP addresses and is active on 72% of days (label B). The results table lists the 20 most similar usernames (5 shown here), each with its total attacks, top country, top three source IPs, the top three ASNs, and a set of "Why similar?" labels (label D), such as a shared lead source country, comparable volume, or sourcing volume, that explain the match.

Above the results, a shared infrastructure panel (label C) lists IP addresses that appear among the top sources of more than one result, and these addresses are shown in the table. In Fig. 4, thirty seven addresses are shared. Each line of the panel names an address and counts how many of the 100 results it is a top source for, followed by examples of those usernames. Two of the most shared addresses (boxed in Fig. 4), 152.89.196.123 and 152.89.196.220, lie in the same /24 network and are top sources for 34 and 33 results respectively, including `manager`, `daniel`, and `oscar`. The panel applies the same analysis to ASes: 13 are shared, led by DigitalOcean (99 of 100 results), Tencent (90), and OVH (40). The panel turns behavioral resemblance into infrastructural evidence: usernames tried by the same hosts are likely drawn from the same credential list and belong to the same campaign.

2) IP Address Similarity: We characterize each IP address by a seventeen-dimensional profile drawn entirely from its activity during the data period. The features fall into four groups: volume (log-scaled totals, daily means, and peaks); temporal shape (the fraction of days on which the address was active, its burst intensity, its largest day-over-day swings, a fitted trend slope, its activity span, and the share of its traffic in the final week); targeting behavior (the number of distinct usernames it tries, how stable that set is from one day to the next, its rotation rate, and the share of attempts taken by its most-tried username); and two flags indicating whether the address belongs to a cloud provider and whether it originates from a major source country. We cluster these profiles with k-means, label each cluster from its centroid, and answer queries in the same way as for usernames.

In Fig. 5, an operator searches for 61.177.173.46 (label A), which falls in Cluster 5, labeled Low-Volume Persistent Exploratory, containing 4,405 addresses. The most similar results are shown in label B, with similarity scores between 91.2% and 80.4%, all belonging to CHINANET-BACKBONE. Behavioral features alone therefore recover infrastructure relationships, suggesting that these hosts are operated together.

Figs. 4 and 5 show how SSHVisual lets operators assemble sets of similar attackers along the username and IP dimensions and see why those attackers were grouped, fulfilling R1.

3) Shared Tagging: Figs. 6–8 show how an operator records a finding so that other operators can see work that has already been done. Tags are organized in a tag repository for each dimension. The repository displays the current operator's name, and every tag and tag application records its author.

In Fig. 6 the operator creates a tag, "similarity to james" (label A), for the username cluster found in Fig. 4. The

Rank	Username	Total Attacks	Avg Daily	Persistence	Max Absolute Δ [1]	Max % Δ [2]	Recent (7d)	First Seen	Last Seen	Max Daily
1	karaf	77.6K	5.2K	21.70% (15d)	32.4K	360.52K%	120	Nov 2, 2022	Jan 8, 2023	44.8K
2	ossuser	82.6K	2.6K	46.40% (32d)	30.4K	276.15K%	690	Nov 3, 2022	Jan 8, 2023	46.0K
3	snow	248.2K	5.2K	69.60% (48d)	22.0K	168.87K%	41.8K	Nov 1, 2022	Jan 8, 2023	35.2K
4	nmrsu	89.4K	1.8K	72.50% (50d)	35.0K	145.91K%	2.9K	Nov 5, 2022	Jan 8, 2023	43.8K
5	lenovo	25.3K	476.51	76.80% (53d)	8.5K	283.10K%	5.7K	Nov 1, 2022	Jan 8, 2023	8.5K
6	fb	9.7K	225.67	62.30% (43d)	4.7K	470.10K%	403	Nov 5, 2022	Jan 8, 2023	4.7K
7	inspur	40.4K	684.71	85.50% (59d)	5.3K	263.05K%	11.2K	Nov 1, 2022	Jan 8, 2023	5.3K
8	radware	12.3K	686.11	26.10% (18d)	4.8K	241.25K%	1.9K	Nov 1, 2022	Jan 8, 2023	4.8K

Fig. 3. Usernames tab sorted by Max Absolute Δ (A) and Max % Δ (B), surfacing bursty usernames (C).

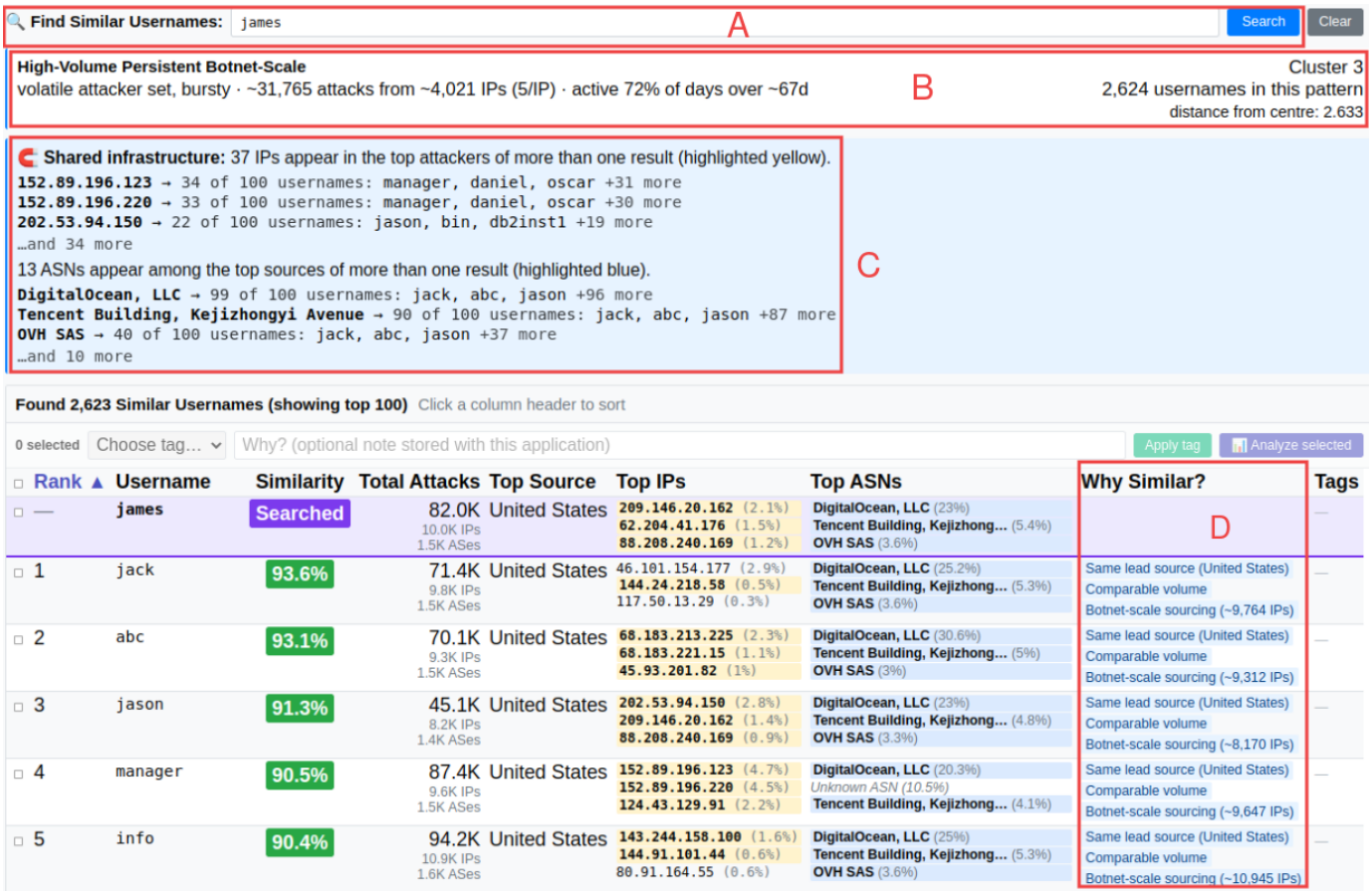


Fig. 4. Similarity search for username `james` (A). The cluster summary (B), shared infrastructure panel (C), and "Why similar?" (D) labels explain why the results were grouped with the query.

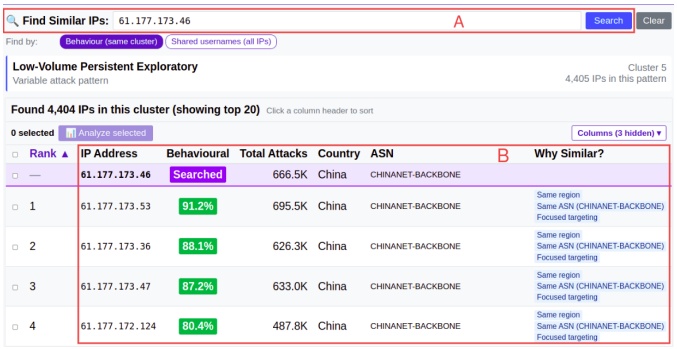


Fig. 5. Similarity search (A) for IP 61.177.173.46. Nearest neighbors are adjacent addresses on the same network (B).

repository now lists the tag with its description (label B) and its author (label C). Tags are applied from the similarity results. In Fig. 7 the operator selects seven usernames (label A), chooses the new tag, and adds a note explaining this particular application: "recurring ASNs" (label B). In Fig. 8 the repository reports seven applications (labels A and F). Filtering the list by tag shows each tagged username with its note (labels B and C), the operator who applied it (label D), and a timestamp (label E), and any application can be removed with Untag. Tagged usernames also appear in the Tags column of subsequent similarity searches, so an operator who later retrieves the same usernames sees immediately that they have already been investigated.

Separating a tag's meaning from the rationale for each application lets a later operator understand both what a group represents and why each member belongs to it, without repeating the analysis. The recorded author and timestamp indicate whether a finding is recent. Figs. 6–8 thus show how tagging supports collaboration among operators, fulfilling R3.

V. Evaluation

We evaluate SSHVisual on its ability to uncover patterns. We show this in two ways: 1) finding campaigns in which the same list of attempted usernames recurs across IPs in different countries and 2) finding entities to block that act alike but would be buried underneath entities that dominate attacks. Both the subsections start with a network operator coming across a single IP, for example from an intrusion detection alert, an abuse report, a flag report on stackoverflow or the discovery tables in SSHVisual. On its own, such an IP is one address among tens of thousands; SSHVisual helps the operator find what else it is connected to.

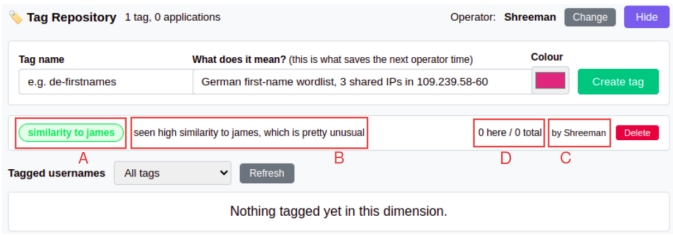


Fig. 6. A newly created tag (A), listed with its description (B), author (C), and application counts (D).

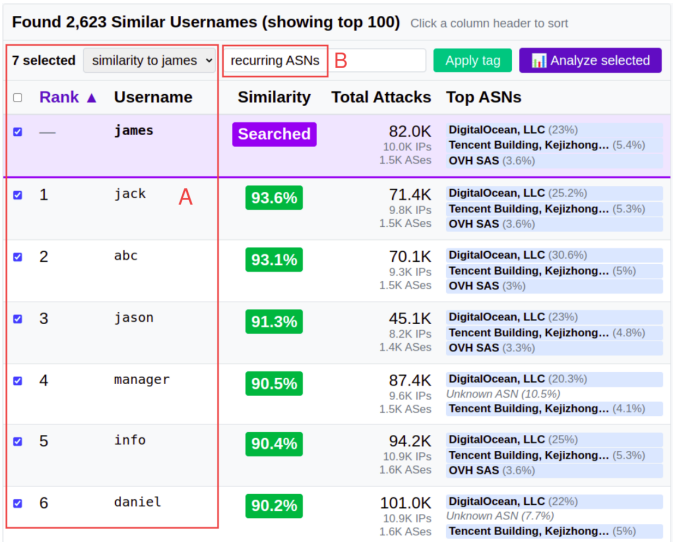


Fig. 7. Applying the tag to seven usernames from the similarity results (A), with a note explaining the application (B).

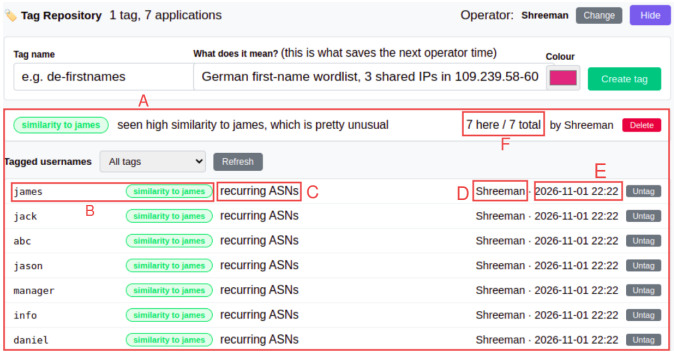


Fig. 8. The tag repository after the tag is applied (A), listing each tagged username (B) with its note (C), author (D), timestamp (E), and application counts (F).

A. Username Campaigns

Fig. 9 shows the username search for IP 68.183.213.225 (DigitalOcean, Germany). Four IPs score 100% username similarity with it (labels A–E): 143.244.183.240, 24.199.108.32 and 146.190.113.228 in the United States, and 134.122.62.125 in the Netherlands (not shown in Fig. 9 but at least 12 IPs have a username similarity score greater than 84%). All five try the same 77 usernames in near-identical proportions, and all 16 of

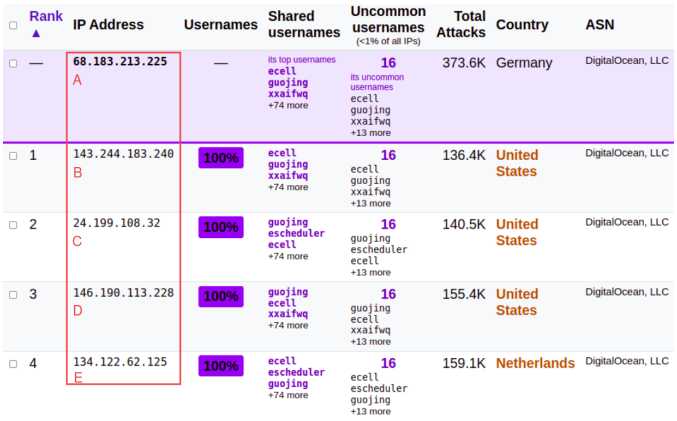


Fig. 9. 100% usernames shared among five IPs. IPs A, B, C, D, E match the IPs A, B, C, D, E in Fig. 10.

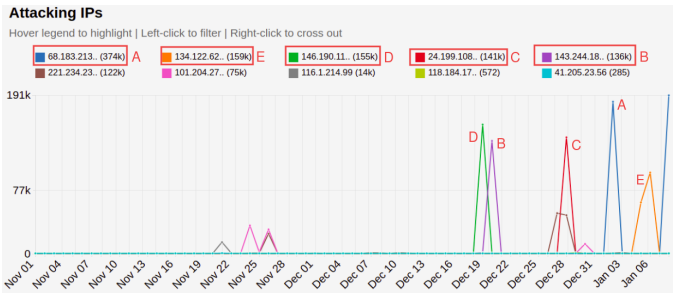


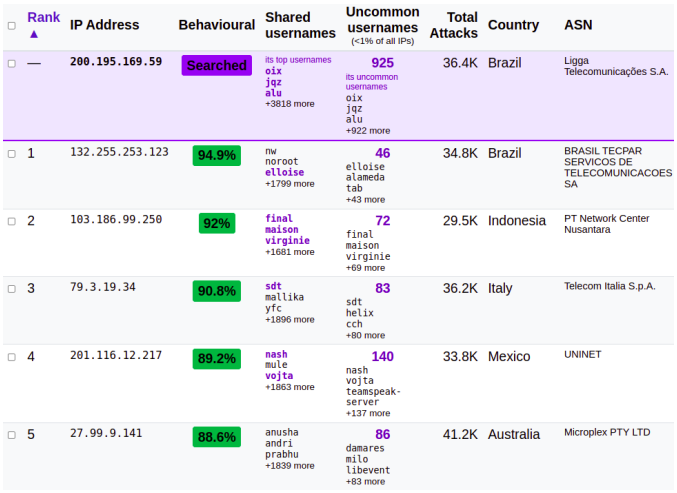
Fig. 10. Five IPs labeled A–E from Fig. 9 in the linechart, along with other IPs with almost the same username list.

the searched IP's uncommon usernames (such as ecell, guojing and xxaifwq) appear in every list. The uncommon usernames matter here: a 100% score built on common usernames like root and admin would mean little, but 16 shared uncommon usernames are unlikely to occur by chance. The five IPs sit on one ASN across three countries (not shown in Fig. 9 but the 12 IPs mentioned before come from 5 countries and 6 ASNs), which is consistent with a single operator renting servers in several regions.

Fig. 10 shows why this campaign is hard to see from volume alone. Each of the IPs is active for only a day or two, and at different times: 146.190.113.228 on Dec 19 (label D), 143.244.183.240 on Dec 20 (label B), 24.199.108.32 on Dec 28 (label C), 68.183.213.225 on Jan 2 and Jan 8 (label A), and 134.122.62.125 on Jan 5 to 6 (label E). The lines never rise together, so the chart suggests five unrelated bursts. Only the shared username list links them into one campaign.

B. Finding Hidden Entities to Block

Fig. 11 shows the IP similarity search for 200.195.169.59, an IP from Ligga Telecomunicações S.A. in Brazil with 36.4K attacks. The five closest matches (88.6% to 94.9% behavioral similarity) come from Brazil, Indonesia, Italy, Mexico and Australia, each from a different ASN (not shown in Fig. 11 but the top 20 similarity matches—at least 86%—to this IP come from 15 different countries and 21 different ASNs). Despite



Rank ▲	IP Address	Behavioural	Shared usernames	Uncommon usernames (<1% of all IPs)	Total Attacks	Country	ASN
—	200.195.169.59	Searched	its top usernames oix jqz alu +3818 more	925 its uncommon usernames oix jqz alu +922 more	36.4K	Brazil	Ligga Telecomunicações S.A.
1	132.255.253.123	94.9%	nw noroot elloise +1799 more	46 elloise alameda tab +43 more	34.8K	Brazil	BRASIL TECPAR SERVICOS DE TELECOMUNICACOES SA
2	103.186.99.250	92%	final maison virginie +1681 more	72 final maison virginie +69 more	29.5K	Indonesia	PT Network Center Nusantara
3	79.3.19.34	90.8%	sdt mallika yfc +1896 more	83 sdt helix cch +80 more	36.2K	Italy	Telecom Italia S.p.A.
4	201.116.12.217	89.2%	nash mule vojta +1863 more	140 nash vojta teamspeak-server +137 more	33.8K	Mexico	UNINET
5	27.99.9.141	88.6%	anusha andri prabhu +1839 more	86 damares milo libevent +83 more	41.2K	Australia	Microplex PTY LTD

Fig. 11. Five IPs that behave like 200.195.169.59.

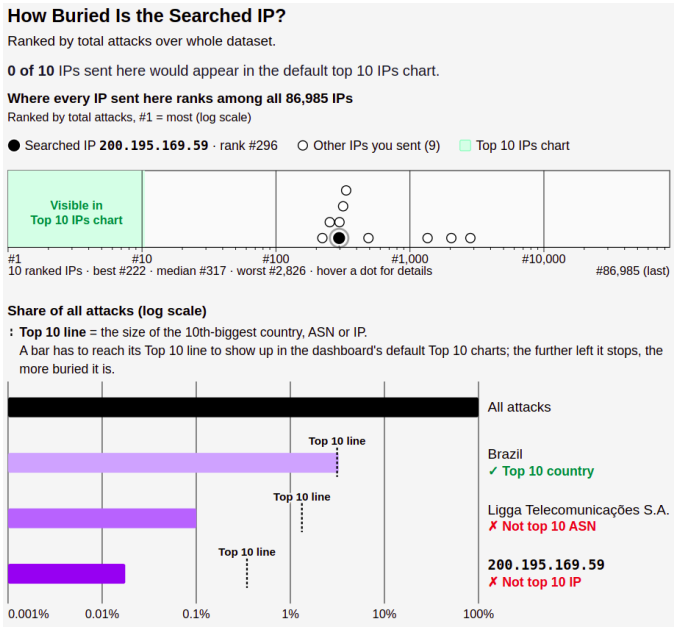


Fig. 12. How buried 200.195.169.59 and its similar IPs are.

this spread, they behave alike: each made roughly 30K to 41K attacks, is tagged as exploratory targeting (that is, they use many different usernames), and shares between 1,684 and 1,899 usernames with the searched IP, 46 to 140 of which are uncommon (tried by fewer than 1% of all IPs). The searched IP itself tried 3,821 usernames, 925 of them uncommon. This is the pattern: low-volume IPs working through large and similarly uncommon username lists from many countries at once.

Fig. 12 shows how buried these IPs are despite their interesting nature to administrators. None of the 10 IPs (selected from Fig. 11) would appear in the top volume IPs chart. The searched IP ranks #296 of 86,985 IPs and accounts for 0.018% of 206.6M attacks. Its country, Brazil, is large enough to appear in the top 10 countries (#8), but its ASN is 13 times too small to appear in the top 10 ASNs, and the IP itself is 20 times too small to appear in the top 10 IPs. The other IPs rank between #222 and #2,826. An administrator looking only at the high volume charts would see Brazil but never these IPs. With SSHVisual, administrators know not just the IP they found to block but also similar IPs that they should block, and most importantly, why they should block them.

VI. Conclusion

We presented SSHVisual, a visual analytics system that treats the username as a first-class analytic dimension alongside the IP address, AS, and country in SSH attack data. SSHVisual pairs a discovery table that ranks every entity with coordinated linecharts that follow Shneiderman's mantra, so operators can move between precise lookup and temporal behavior. Similarity search groups attackers by behavior along the username and IP dimensions and explains each match. Shared tagging preserves those groups so that findings persist and remain visible across operators. Our case studies show that this combination surfaces patterns that volume-based, IP-centric views miss. Low-volume IPs spread across many countries and ASes worked through similar lists of uncommon usernames, yet none ranked high enough to appear in any top-ten chart. IPs that never attacked on the same day were linked into a single campaign only by the 77 usernames they shared. Because SSH honeypot traffic offers no normal baseline for automated detection, these results suggest that exploratory, operator-driven analysis remains a necessary complement to automated methods, and that the username is a valuable lens for recognizing coordinated attackers.

Appendix A
Discovery Table Columns

Table I lists every column of the discovery table (Section IV-A1) and the tabs in which it appears.

TABLE I
DISCOVERY TABLE COLUMNS BY TAB

Column	Countries	IPs	ASNs	Usernames
Total attacks	✓	✓	✓	✓
Average daily attacks	✓	✓	✓	✓
Persistence	✓	✓	✓	✓
Max absolute Δ	✓	✓	✓	✓
Max % Δ	✓	✓	✓	✓
Attacks in last 7 days	✓	✓	✓	✓
First / last seen	✓	✓	✓	✓
Max daily count	✓	✓	✓	✓
Trend	✓	✓	✓	✓
Burst intensity	✓	✓	✓	✓
Unique usernames	✓	✓	✓	
Username stability	✓	✓	✓	
Username concentration	✓	✓	✓	
Unique IPs	✓		✓	✓
IP stability	✓		✓	✓
IP concentration	✓		✓	✓
Unique ASNs	✓			✓
ASN stability	✓			✓
ASN concentration	✓			✓
Unique countries			✓	✓
Country stability			✓	✓
Country concentration			✓	✓
ASN name, Country		✓		
Primary country			✓	
Behavior badge	✓			

REFERENCES

[1] T. Ylonen, "Ssh–secure login connections over the internet," in *Proceedings of the 6th USENIX Security Symposium*, vol. 37, 1996, pp. 40–52.

[2] S. K. Singh, S. Gautam, C. Cartier, S. Patil, and R. Ricci, "Where the wild things are: Brute-Force SSH attacks in the wild and how to stop them," in *21st USENIX Symposium on Networked Systems Design and Implementation (NSDI 24)*. Santa Clara, CA: USENIX Association, Apr. 2024, pp. 1731–1750. [Online]. Available: https://www.usenix.org/conference/nsdi24/presentation/singh-sachin

[3] M. Antonakakis, T. April, M. Bailey, M. Bernhard, E. Bursztein, J. Cochran, Z. Durumeric, J. A. Halderman, L. Invernizzi, M. Kallitsis *et al.*, "Understanding the mirai botnet," in *26th USENIX security symposium (USENIX Security 17)*, 2017, pp. 1093–1110.

[4] C. Kolias, G. Kambourakis, A. Stavrou, and J. Voas, "Ddos in the iot: Mirai and other botnets," *computer*, vol. 50, no. 7, pp. 80–84, 2017.

[5] F. Dang, Z. Li, Y. Liu, E. Zhai, Q. A. Chen, T. Xu, Y. Chen, and J. Yang, "Understanding fileless attacks on linux-based iot devices with honeycloud," in *Proceedings of the 17th Annual International Conference on Mobile Systems, Applications, and Services*, 2019, pp. 482–493.

[6] S. Pastrana and G. Suarez-Tangil, "A first look at the crypto-mining malware ecosystem: A decade of unrestricted wealth," in *Proceedings of the Internet Measurement Conference*, 2019, pp. 73–86.

[7] V. T. Guimarães, C. M. D. S. Freitas, R. Sadre, L. M. R. Tarouco, and L. Z. Granville, "A survey on information visualization for network and service management," *IEEE Communications Surveys & Tutorials*, vol. 18, no. 1, pp. 285–323, 2016.

[8] H. Shiravi, A. Shiravi, and A. A. Ghorbani, "A survey of visualization systems for network security," *IEEE Transactions on Visualization and Computer Graphics*, vol. 18, no. 8, pp. 1313–1329, 2012.

[9] T. Zhang, X. Wang, Z. Li, F. Guo, Y. Ma, and W. Chen, "A survey of network anomaly visualization," *Science China Information Sciences*, vol. 60, no. 12, p. 121101, 2017.

[10] J. Raynor, T. Crnovrsanin, S. Di Bartolomeo, L. South, D. Saffo, and C. Dunne, "The state of the art in bgp visualization tools: A mapping of visualization techniques to cyberattack types," *IEEE Transactions on Visualization and Computer Graphics*, vol. 29, no. 1, pp. 1059–1069, 2022.

[11] C. Wu, S. Sheng, and X. Dong, "Research on visualization systems for ddos attack detection," in *2018 IEEE International Conference on Systems, Man, and Cybernetics (SMC)*. IEEE, 2018, pp. 2986–2991.

[12] C. Gates and C. Taylor, "Challenging the anomaly detection paradigm: A provocative discussion," in *Proceedings of the 2006 workshop on New security paradigms*, 2006, pp. 21–29.

[13] F. Mansmann and S. Mansmann, "Interactive exploration of data traffic with hierarchical network maps," 2006.

[14] J. Oberheide, M. Goff, and M. Karir, "Flamingo: Visualizing internet traffic," in *2006 IEEE/IFIP Network Operations and Management Symposium NOMS 2006*. IEEE, 2006, pp. 150–161.

[15] F. Mansmann, F. Fischer, D. A. Keim, and S. C. North, "Visual support for analyzing network traffic and intrusion detection events using treemap and graph representations," in *Proceedings of the Symposium on Computer Human Interaction for the Management of Information Technology*, 2009, pp. 19–28.

[16] T. Taylor, D. Paterson, J. Glanfield, C. Gates, S. Brooks, and J. McHugh, "Flovis: Flow visualization system," in *2009 Cybersecurity Applications & Technology Conference for Homeland Security*. IEEE, 2009, pp. 186–198.

[17] S. Chen, C. Guo, X. Yuan, F. Merkle, H. Schaefer, and T. Ertl, "Oceans: Online collaborative explorative analysis on network security," in *Proceedings of the Eleventh Workshop on Visualization for Cyber Security*, 2014, pp. 1–8.

[18] T. Tremel, J. Kogel, F. Jauernig, S. Meier, D. Thom, F. Becker, C. Muller, and S. Koch, "Vitalflow: Visual interactive traffic analysis with netflow," in *NOMS 2022-2022 IEEE/IFIP Network Operations and Management Symposium*. IEEE, 2022, pp. 1–6.

[19] L. Xiao, J. Gerth, and P. Hanrahan, "Enhancing visual analysis of network traffic using a knowledge representation," in *2006 IEEE symposium on visual analytics science and technology*. IEEE, 2006, pp. 107–114.

[20] J. R. Goodall and M. Sowul, "Viassist: Visual analytics for cyber defense," in *2009 IEEE conference on technologies for homeland security*. IEEE, 2009, pp. 143–150.

[21] D. H. Jeong, J.-H. Cho, F. Chen, L. Kaplan, A. Jøsang, and S.-Y. Ji, "Interactive web-based visual analysis on network traffic data," *Information*, vol. 14, no. 1, p. 16, 2022.

[22] H. Choi, H. Lee, and H. Kim, "Fast detection and visualization of network attacks on parallel coordinates," *computers & security*, vol. 28, no. 5, pp. 276–288, 2009.

[23] Y. Livnat, J. Agutter, S. Moon, and S. Foresti, "Visual correlation for situational awareness," in *IEEE Symposium on Information Visualization, 2005. INFOVIS 2005.* IEEE, 2005, pp. 95–102.

[24] C. Humphries, N. Prigent, C. Bidan, and F. Majorczyk, "Elvis: Extensible log visualization," in *Proceedings of the Tenth Workshop on Visualization for Cyber Security*, 2013, pp. 9–16.

[25] T. Yu, R. Lippmann, J. Riordan, and S. Boyer, "Ember: a global perspective on extreme malicious behavior," in *Proceedings of the seventh international symposium on visualization for cyber security*, 2010, pp. 1–12.

[26] D. H. Jeong, B.-K. Jeong, and S.-Y. Ji, "Multi-resolution analysis with visualization to determine network attack patterns," *Applied Sciences*, vol. 13, no. 6, p. 3792, 2023.

[27] A. King, B. Huffaker, A. Dainotti, and K. Claffy, "A coordinated view of the temporal evolution of large-scale internet events," *Computing*, vol. 96, no. 1, pp. 53–65, 2014.

[28] D. Ceneda, M. Di Bartolomeo, V. Di Donato, M. Patrignani, M. Pizzonia, and M. Rimondini, "Routingwatch: Visual exploration and analysis of routing events," in *NOMS 2016-2016 IEEE/IFIP Network Operations and Management Symposium*. IEEE, 2016, pp. 591–597.

[29] J. C. Roberts, "State of the art: Coordinated & multiple views in exploratory visualization," in *Fifth international conference on coordinated and multiple views in exploratory visualization (CMV 2007)*. IEEE, 2007, pp. 61–71.

[30] J. S. Yi, Y.-a. Kang, J. T. Stasko, and J. A. Jacko, "Understanding and characterizing insights: how do people gain insights using information visualization?" in *Proceedings of the 2008 Workshop on BEyond time and errors: novel evaLuation methods for Information Visualization*, 2008, pp. 1–6.

[31] R. Djap, C. Lim, K. E. Silaen, and A. Yusuf, "Xb-pot: Revealing honeypot-based attacker's behaviors," in *2021 9th International Conference on Information and Communication Technology (ICoICT)*. IEEE, 2021, pp. 550–555.

[32] V. Visoottiviseth, U. Jaralrungroj, E. Phoomrungraungsuk, and P. Kultanon, "Distributed honeypot log management and visualization of attacker geographical distribution," in *2011 Eighth International Joint Conference on Computer Science and Software Engineering (JCSSE)*. IEEE, 2011, pp. 23–28.

[33] M. Baykara and R. Das, "A novel honeypot based security approach for real-time intrusion detection and prevention systems," *Journal of Information Security and Applications*, vol. 41, pp. 103–116, 2018.

[34] A. Belqruch and A. Maach, "Scada security using ssh honeypot," in *Proceedings of the 2nd International Conference on Networking, Information Systems & Security*, 2019, pp. 1–5.

[35] I. Koniaris, G. Papadimitriou, and P. Nicopolitidis, "Analysis and visualization of ssh attacks using honeypots," in *Eurocon 2013*. IEEE, 2013, pp. 65–72.

[36] A. Kyriakou and N. Sklavos, "Container-based honeypot deployment for the analysis of malicious activity," in *2018 Global Information Infrastructure and Networking Symposium (GIIS)*. IEEE, 2018, pp. 1–4.

[37] I. Koniaris, G. Papadimitriou, P. Nicopolitidis, and M. Obaidat, "Honeypots deployment for the analysis and visualization of malware activity and malicious connections," in *2014 IEEE international conference on communications (ICC)*. IEEE, 2014, pp. 1819–1824.

[38] D. Fraunholz, M. Zimmermann, S. D. Anton, J. Schneider, and H. D. Schotten, "Distributed and highly-scalable wan network attack sensing and sophisticated analysing framework based on honeypot technology," in *2017 7th International Conference on Cloud Computing, Data Science & Engineering-Confluence*. IEEE, 2017, pp. 416–421.

[39] J. R. Goodall, E. D. Ragan, C. A. Steed, J. W. Reed, G. D. Richardson, K. M. Huffer, R. A. Bridges, and J. A. Laska, "Situ: Identifying and explaining suspicious behavior in networks," *IEEE transactions on visualization and computer graphics*, vol. 25, no. 1, pp. 204–214, 2018.

[40] Q. Liao and A. Striegel, "Intelligent network management using graph differential anomaly visualization," in *2012 IEEE Network Operations and Management Symposium*. IEEE, 2012, pp. 1008–1014.

[41] W. A. Pike, C. Scherrer, and S. Zabriskie, "Putting security in context: Visual correlation of network activity with real-world information," in *VizSEC 2007: Proceedings of the Workshop on Visualization for Computer Security*. Springer, 2008, pp. 203–220.

[42] L. Hao, C. G. Healey, and S. E. Hutchinson, "Ensemble visualization for cyber situation awareness of network security data," in *2015 IEEE Symposium on Visualization for Cyber Security (VizSec)*. IEEE, 2015, pp. 1–8.

[43] F. Pouget, G. Urvoy-Keller, and M. Dacier, "Time signatures to detect multi-headed stealthy attack tools," in *18th Annual FIRST Conference*, Baltimore, MD, USA, 2006.

[44] O. Thonnard and M. Dacier, "A framework for attack patterns' discovery in honeynet data," *Digital Investigation*, vol. 5, pp. S128–S139, 2008, proc. DFRWS 2008.

[45] R. Sommer and V. Paxson, "Outside the closed world: On using machine learning for network intrusion detection," in *2010 IEEE symposium on security and privacy*. IEEE, 2010, pp. 305–316.

[46] S. Axelsson, "The base-rate fallacy and the difficulty of intrusion detection," *ACM Transactions on Information and System Security (TISSEC)*, vol. 3, no. 3, pp. 186–205, 2000.

[47] D. M. Best, R. P. Hafen, B. K. Olsen, and W. A. Pike, "Atypical behavior identification in large-scale network traffic," in *2011 IEEE Symposium on Large Data Analysis and Visualization*. IEEE, 2011, pp. 15–22.

[48] Y. Xie, F. Yu, K. Achan, E. Gillum, M. Goldszmidt, and T. Wobber, "How dynamic are ip addresses?" in *Proceedings of the 2007 conference on Applications, technologies, architectures, and protocols for computer communications*, 2007, pp. 301–312.

[49] B. Stone-Gross, M. Cova, L. Cavallaro, B. Gilbert, M. Szydlowski, R. Kemmerer, C. Kruegel, and G. Vigna, "Your botnet is my botnet: analysis of a botnet takeover," in *Proceedings of the 16th ACM conference on Computer and communications security*, 2009, pp. 635–647.

[50] G. C. Moura, C. Ganán, Q. Lone, P. Poursaied, H. Asghari, and M. van Eeten, "How dynamic is the isps address space? towards internet-wide dhcp churn estimation," in *2015 IFIP Networking Conference (IFIP Networking)*. IEEE, 2015, pp. 1–9.

[51] J. L. Guerra, E. Veas, and C. A. Catania, "A study on labeling network hostile behavior with intelligent interactive tools," in *2019 IEEE Symposium on Visualization for Cyber Security (VizSec)*. IEEE, 2019, pp. 1–10.

[52] J. Owens and J. Matthews, "A study of passwords and methods used in brute-force SSH attacks," in *USENIX Workshop on Large-Scale Exploits and Emergent Threats (LEET)*, San Francisco, CA, USA, 2008. [Online]. Available: https://lin-web.clarkson.edu/~jmatthew/publications/leet08.pdf

[53] Y. Wu, P. M. Cao, A. Withers, Z. T. Kalbarczyk, and R. K. Iyer, "Mining threat intelligence from billion-scale SSH brute-force attacks," in *Workshop on Decentralized IoT Systems and Security (DISS)*. San Diego, CA, USA: Internet Society, 2020. [Online]. Available: https://www.ndss-symposium.org/wp-content/uploads/2020/04/diss2020-23007-paper.pdf

[54] B. Shneiderman, "The eyes have it: A task by data type taxonomy for information visualizations," in *Proceedings of the IEEE Symposium on Visual Languages*, 1996, pp. 336–343.

[55] W. Javed, B. McDonnel, and N. Elmqvist, "Graphical perception of multiple time series," *IEEE Transactions on Visualization and Computer Graphics*, vol. 16, no. 6, pp. 927–934, 2010.